Suman Mandal

Durgapur, India — sumanmandal.dev@gmail.com — +91-8240715898
GitHub: github.com/jijo-OO7

Summary

Open Source Security Engineer specializing in Go, Kubernetes, cloud-native security, and software supply chain security. Active contributor to Kubescape, Knative, OpenTelemetry, SPDX, and the Chainguard ecosystem. Designed and implemented production security features including metadata protection, cryptographic workflows, and cloud-native tooling while collaborating directly with maintainers through design discussions, code reviews, testing, documentation, and upstream feature delivery.

Professional Experience

Independent Open Source Security Engineer *2025–Present*

- Develop production-grade security features in Go across CNCF projects with a primary focus on Kubernetes security and cloud-native infrastructure.
- Design, implement, test, document, and maintain production-quality software in collaboration with project maintainers.
- Build secure software emphasizing automation, cryptography, maintainability, testing, and open-source engineering best practices.

Associate Software Engineer – Dietary Business Intelligence (DBI360) *Sep 2024–Oct 2024*

- Worked on backend software development and engineering tasks in an enterprise environment.

Open Source Contributions

Kubescape (CNCF Cloud Security) *2026–Present*
Official Blog: Metadata Protection — Official Documentation: Protecting Report Metadata

- Designed and implemented an end-to-end metadata protection framework spanning deterministic anonymization, encryption, decryption, CLI integration, testing, and documentation for Kubernetes security reports.
- Implemented AES-256-GCM encryption/decryption workflows enabling secure report sharing while maintaining backward compatibility.
- Built report transformation pipelines, secure metadata handling, session anonymization, and resource identifier mapping.
- Collaborated closely with maintainers through architecture discussions, implementation reviews, debugging sessions, documentation, testing, and upstream feature delivery.

OpenTelemetry Collector (Core & Contrib) *2025–Present*

- Contributed Go-based fixes across OpenTelemetry Collector and Collector Contrib improving telemetry pipeline reliability and observability.
- Worked within large distributed Go codebases while collaborating through upstream code reviews and testing workflows.

Knative Project *Since Feb 2026*

- Contributed Go code, tooling improvements, testing enhancements, documentation, and repository maintenance across Knative Eventing and Kafka Broker.
- Collaborated with maintainers on production cloud-native systems and Kubernetes infrastructure.

Additional Open Source Contributions

- **Chainguard** – Contributed improvements to the Chainguard developer website through upstream pull requests.
- **SPDX** – Contributed documentation and tooling improvements supporting the software supply chain ecosystem.

Technical Projects

ChainRisk – Software Supply Chain Risk Analysis Tool (Go)
github.com/jijo-OO7/chainrisk

- Designed and developed a Go CLI for software supply chain risk analysis, modeling dependency graphs and evaluating transitive dependency exposure.
- Implemented a modular architecture supporting future SBOM integration and extensible security analysis.
- Built with idiomatic Go emphasizing maintainability, modularity, and secure software engineering.

Technical Skills

Languages: Go, Python, Java
Cloud Native: Kubernetes, Docker, Helm, Knative
Security: Cloud Security, Container Security, Software Supply Chain Security, Cryptography, AES-256-GCM, Kubescape
Observability: OpenTelemetry, Distributed Systems, Telemetry Pipelines
Development: Linux, Git, GitHub Actions, CI/CD, Unit Testing, Open Source Collaboration

Education

National Institute of Technology Jamshedpur *2021–2024*
Master of Computer Applications (Computer Science) CGPA: 7.57

Techno India Salt Lake *2016–2019*
Bachelor of Computer Applications CGPA: 7.00